Tiny Object Detection with Single Point Supervision

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딥러닝논문읽기모임 7기

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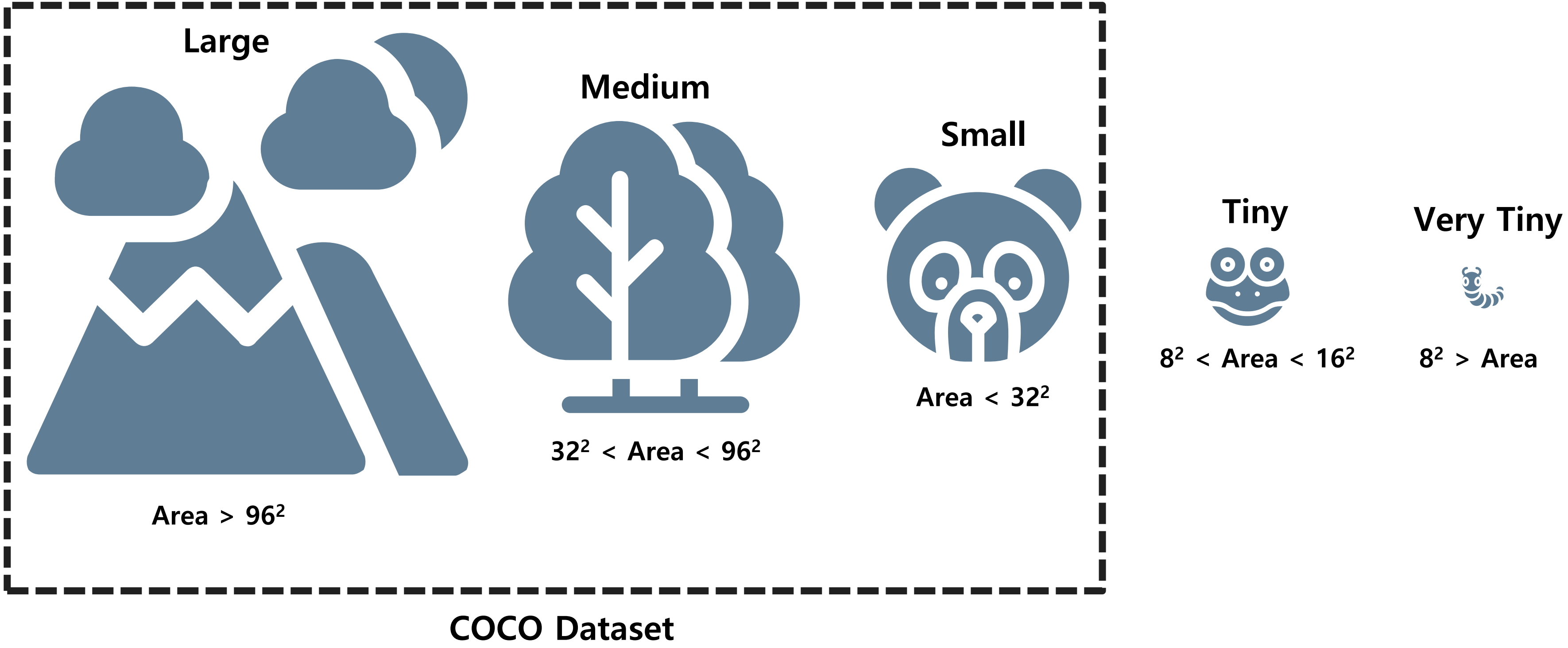
01. Introduction

1. Tiny Objects Detection

- **Size criteria for Object Detection Task**



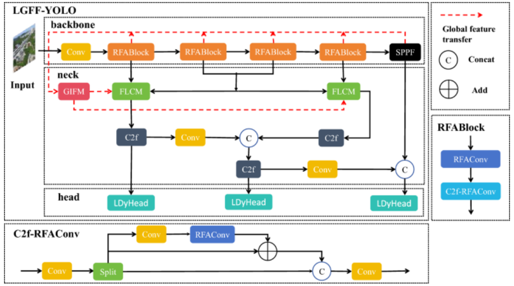
01. Introduction

1. Tiny Objects Detection

- **Previous Works**

Multi-Scale Representation

- Image Level Method / Feature Level Method

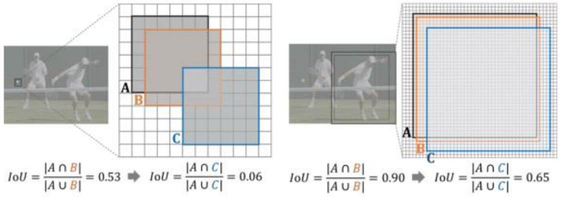


Super Resolution

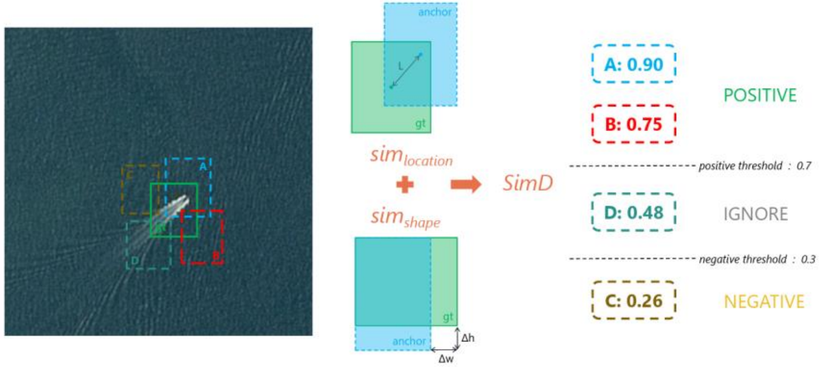
- GAN

Learning Strategies

- Loss Function / Labeling Assignments



(a) Tiny scale object (b) Normal scale object



Peng, Hongxing, et al. "LGFF-YOLO: small object detection method of UAV images based on efficient local–global feature fusion.", 2024
Wang, Jinwang, et al. "A normalized Gaussian Wasserstein distance for tiny object detection.", 2021
Shi, Shuohao, et al. "Similarity distance-based label assignment for tiny object detection.", 2024

01. Introduction

2. Point Supervised Tiny Objects Detection

- **Intuition**

Original Image

(b) Point-annotated Image

Xu. hang, et al. "Detecting tiny objects in aerial images: A normalized Wasserstein distance and a new benchmark.", 2022

01. Introduction

2. Point Supervised Tiny Objects Detection

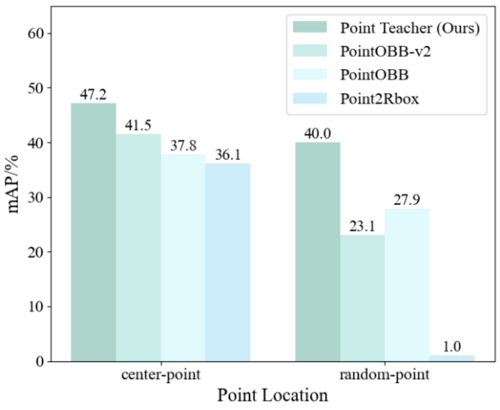
- **Previous Works**

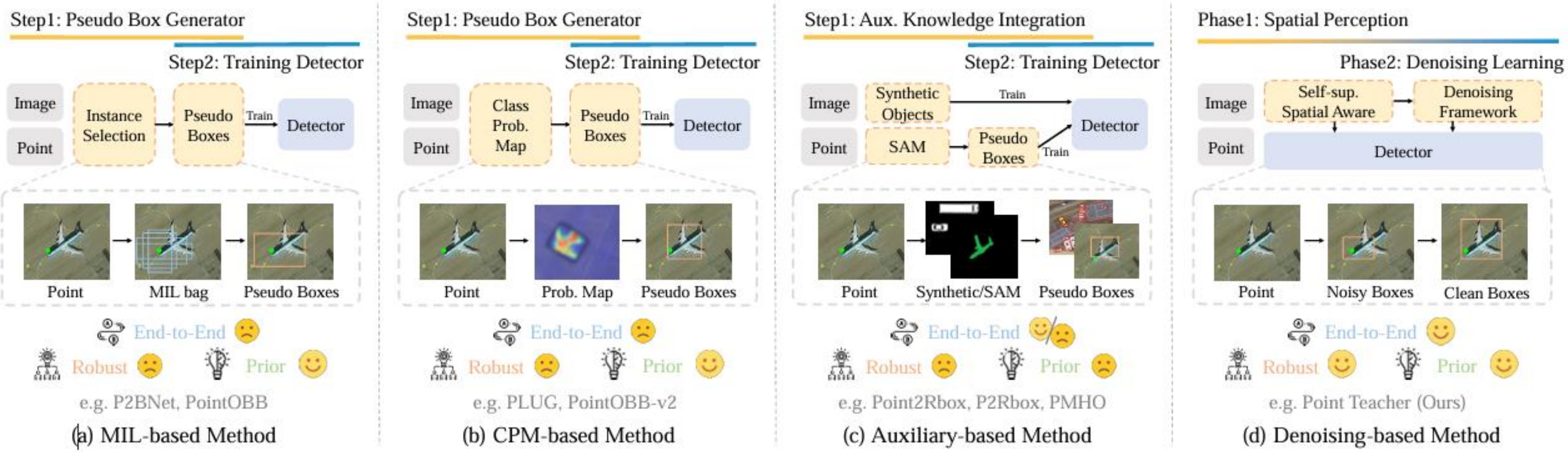
Point-supervised Object Detection (PSOD)

- Center Point / Center Region / Gaussian Region / Mask Region

Point-to-box Training

- Auxiliary Knowledge / Synthetic Object / Pseudo Box / Mask by SAM



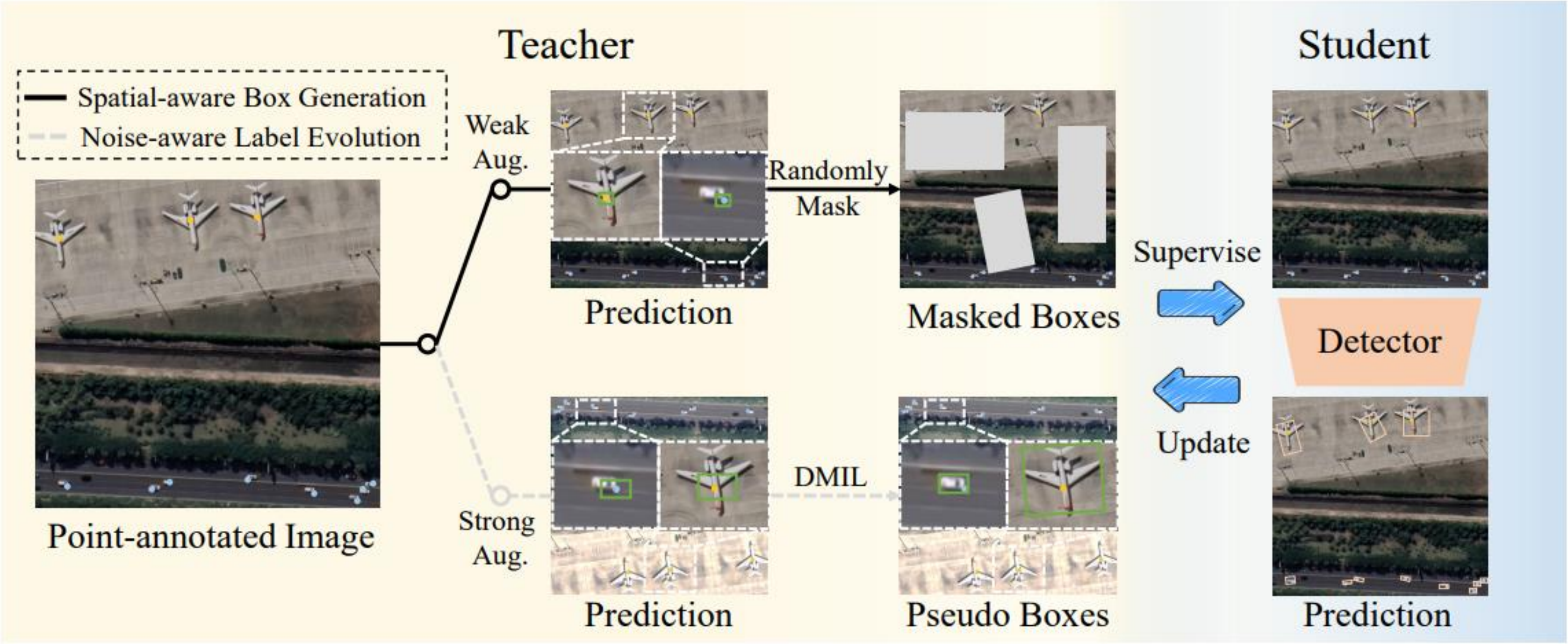


(a) MIL-based Method (b) CPM-based Method (c) Auxiliary-based Method (d) Denoising-based Method

01. Introduction

2. Point Supervised Tiny Objects Detection

- **Point Teacher**



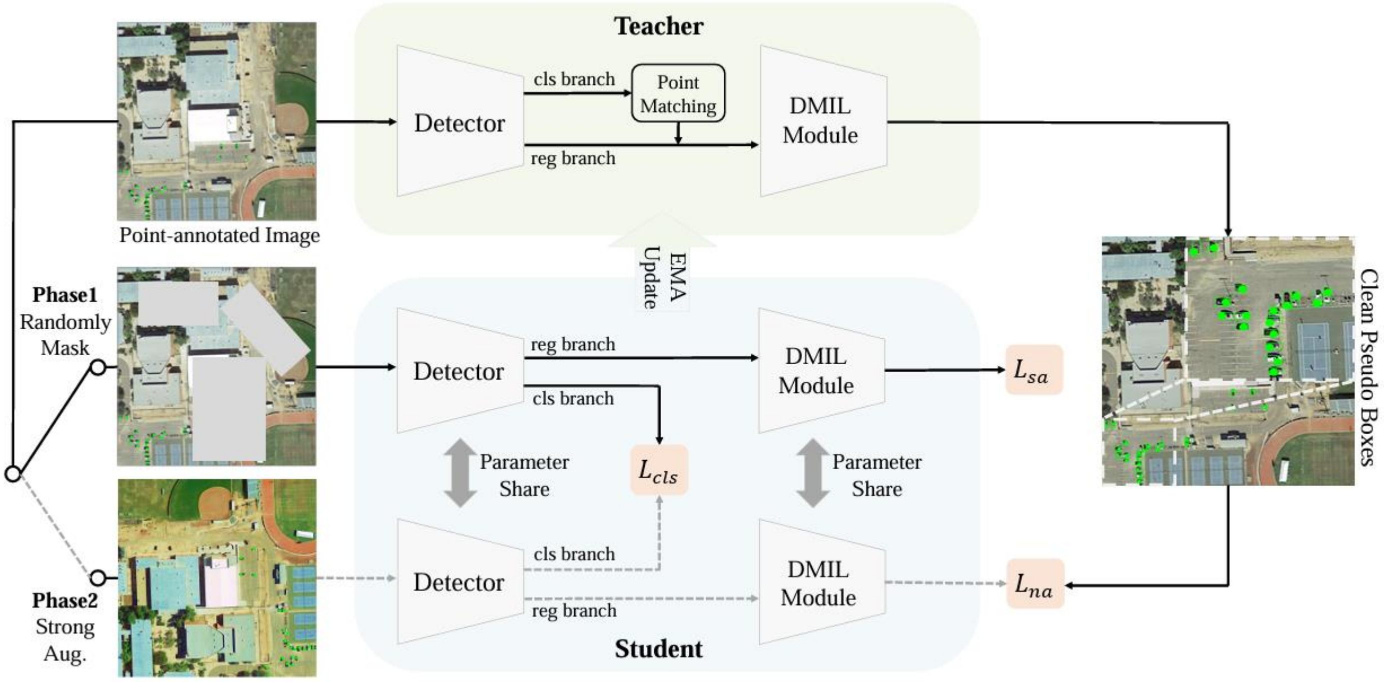
(c) Our Point Teacher Framework

02

Method

02. Method

1. Overview



02. Method

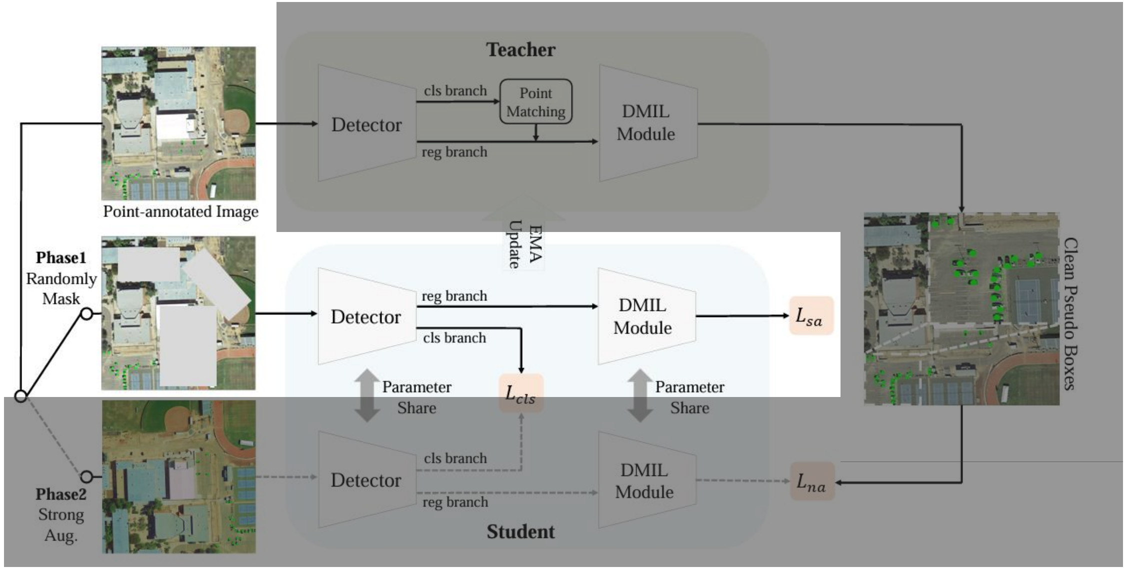
2. Spatial-aware Box Generation

$R(\{cx, cy, w, h\})$: 랜덤하게 선택된 Mask 영역 (N개)

$P(\{cx, cy, w, h\})$: Mask 영역을 네트워크가 Prediction한 결과

$$L_{reg}^{sa} = L_{bbox}(P, R),$$

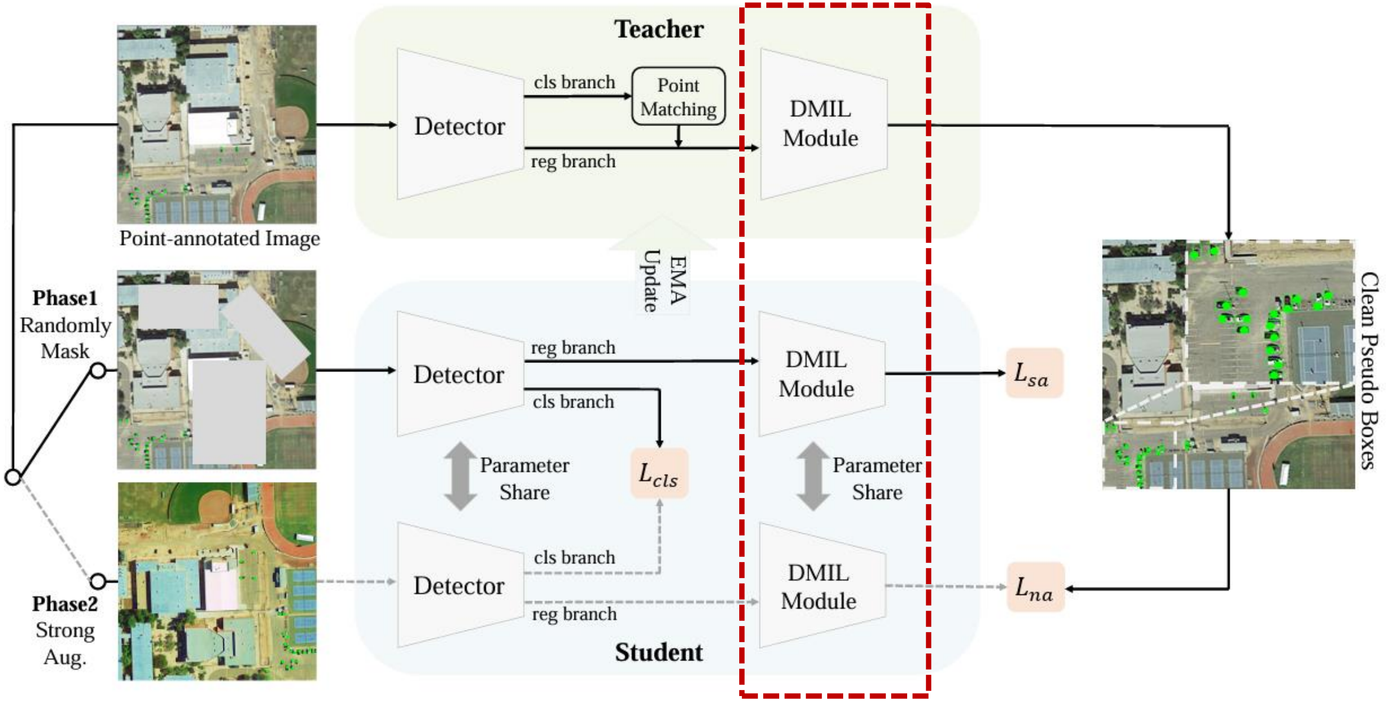
$$L_{sa} = \boxed{L_{reg}^{sa}} + \alpha_1 \cdot L_{reg}^{dmil} + \alpha_2 \cdot L_{cls}^{dmil}$$



02. Method

2. Spatial-aware Box Generation

- **DMIL(Dynamic Multiple Instance Learning) Module**



02. Method

2. Spatial-aware Box Generation

- **DMIL(Dynamic Multiple Instance Learning) Module**

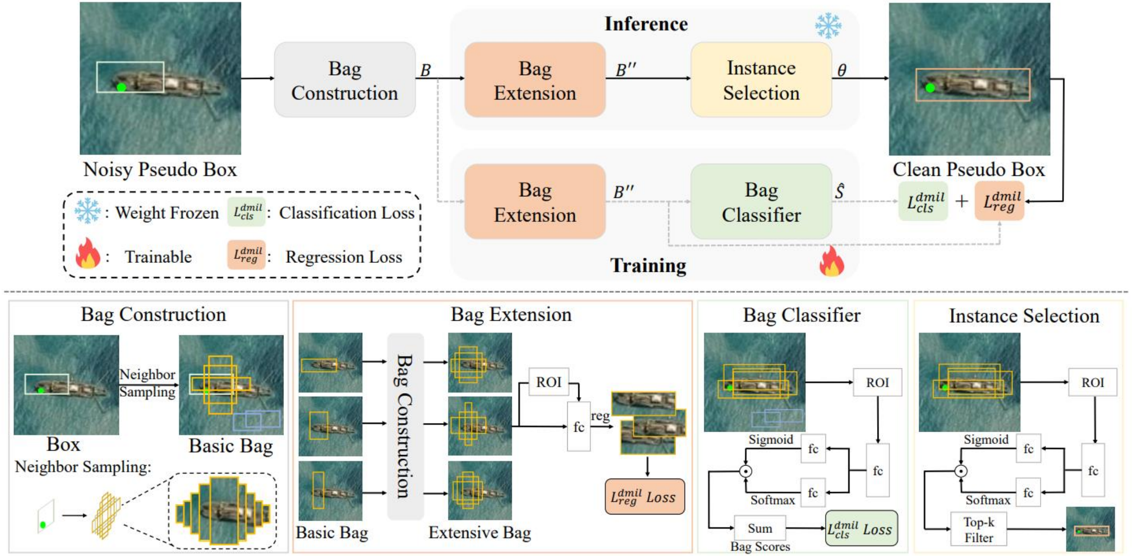


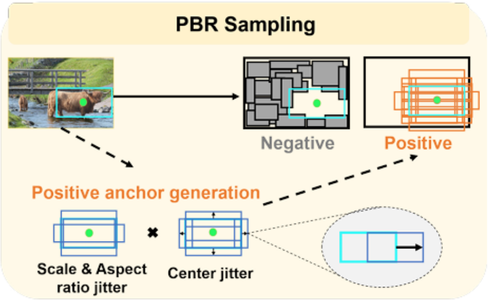
Figure 4: The workflow of Dynamic Multiple Instance Learning Module (DMIL). DMIL comprises four stages: Bag Construction, Bag Extension, Bag Classifier, and Instance Selection. The Bag Construction and Bag Extension stages ensure the creation of high-quality bags. The Bag Classifier is used to train the classification and discrimination capabilities of the bags. Finally, Instance Selection uses the scores from the Bag Classifier to merge instances, generating pseudo boxes.

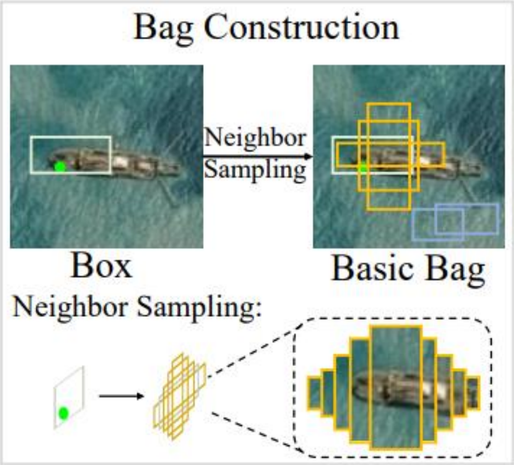
02. Method

2. Spatial-aware Box Generation

- **DMIL(Dynamic Multiple Instance Learning) Module**

Bag Construction





$R(\{cx, cy, w, h\})$: 랜덤하게 선택된 Mask 영역 (N개)

$B_R \in \mathbb{R}^{N \times U_1 \times 4}$: Mask Proposal bags

$Z(\{cx, cy, w, h\})$: Coarse Pseudo Box (M개)

$B_Z \in \mathbb{R}^{M \times U_1 \times 4}$: Pseudo Box Proposal bags

$B_{neg} \in \mathbb{R}^{U_{neg} \times 4}$: Pseudo Box Proposal bags

U_{neg}, U_1 : Bag안에 Proposal 개수

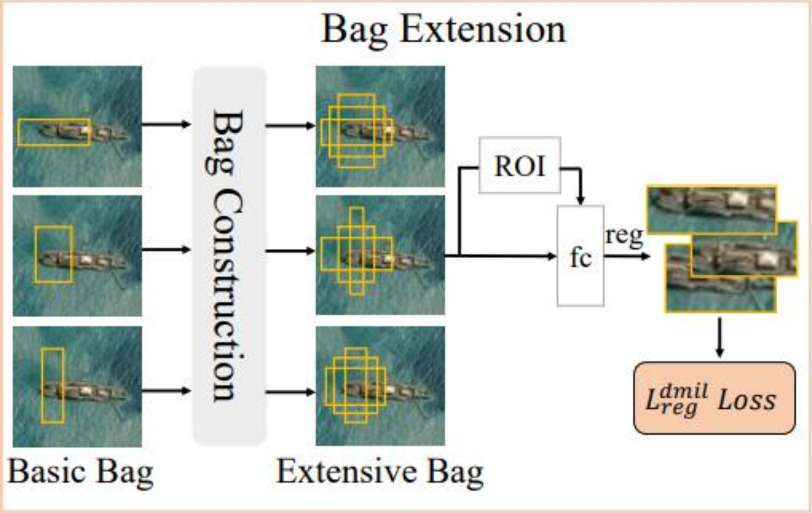
Chen, Pengfei, et al. "Point-to-box network for accurate object detection via single point supervision.", 2022.

02. Method

2. Spatial-aware Box Generation

- **DMIL(Dynamic Multiple Instance Learning) Module**

Bag Extension



$B_R \in \mathbb{R}^{N\times U_1\times 4}$: Mask Proposal bags

$B'_R \in \mathbb{R}^{N\times U_1\times U_2\times 4}$: Expand Bags

$F'_R \in \mathbb{R}^{N\times U_1\times U_2\times D}$: 7 x 7 RoIAlign, FC Layer를 통과한 Feature

$B''_R \in \mathbb{R}^{N\times U_1\times U_2\times 4}$: Refined Bags

$B_Z \in \mathbb{R}^{M\times U_1\times 4}$: Pseudo Box Proposal bags

$B'_Z \in \mathbb{R}^{M\times U_1\times U_2\times 4}$: Expand Bags

$F'_Z \in \mathbb{R}^{N\times U_1\times U_2\times D}$: 7 x 7 RoIAlign, FC Layer를 통과한 Feature

$B''_Z \in \mathbb{R}^{M\times U_1\times U_2\times 4}$: Refined Bags

U_2 : 새로 생성된 proposal 개수

- Spatial-aware Box Generation Stage에서는, Masked Area에 대한 정답값이 존재하므로 Supervise 가능

$R' \in \mathbb{R}^{N\times U_1\times U_2\times 4}$

$$L_{reg}^{dmil} = L_{bbox}(B''_R, R').$$

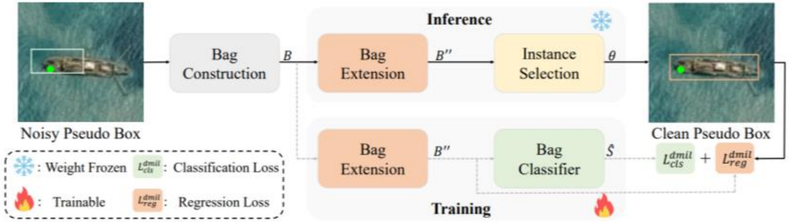
$$L_{sa} = L_{reg}^{sa} + \alpha_1 \cdot \boxed{L_{reg}^{dmil}} + \alpha_2 \cdot L_{cls}^{dmil}$$

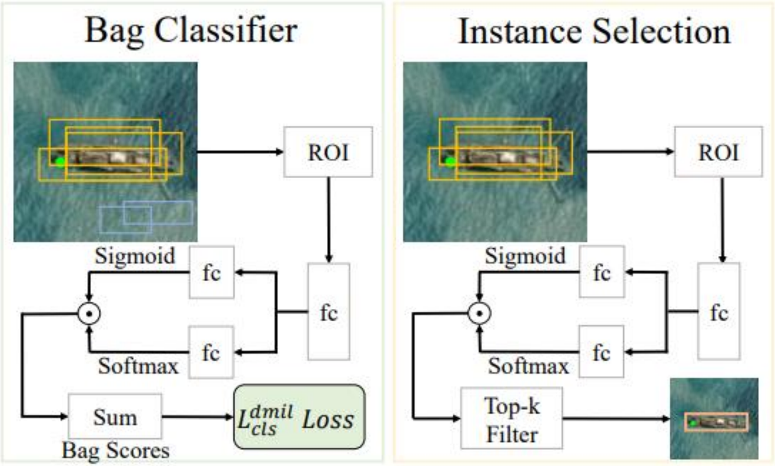
02. Method

2. Spatial-aware Box Generation

- **DMIL(Dynamic Multiple Instance Learning) Module**

Bag Classifier / Instance Selection





$$L_{cls}^{dmil} = L_{cls}(\hat{S}, \xi^{pos}) + L_{cls}(\hat{S}^{neg}, \xi^{neg}),$$

$$L_{sa} = L_{reg}^{sa} + \alpha_1 \cdot L_{reg}^{dmil} + \alpha_2 \cdot \boxed{L_{cls}^{dmil}}$$

$F_Z'' \in \mathbb{R}^{M \times U_1 \times U_2 \times D}$: B_Z''를 RoIAlign, FC Layer를 통해 획득한 Feature

$O^{cls} \in \mathbb{R}^{M \times U_1 \times U_2 \times C}$: F_Z'' 를 Classification Brench를 통과한 결과

S^{cls} : O^{cls} 를 Sigmoid Function에 통과하여 얻은 Score

$O^{ins} \in \mathbb{R}^{M \times U_1 \times U_2 \times C}$: F_Z'' 를 Instance Brench를 통과한 결과

S^{ins} : O^{ins} 를 Softmax Function에 통과하여 얻은 Score

$$O^{cls} = f_{cls}(F_Z''); \quad S_{i,j,k,c}^{cls} = 1/(1 + e^{-O_{i,j,k,c}^{cls}})$$

$$O^{ins} = f_{ins}(F_Z''); \quad S_{i,j,k,c}^{ins} = e^{O_{i,j,k,c}^{ins}} / \sum_{x=1}^{U_2} e^{O_{i,j,x,c}^{ins}}$$

$$S = S^{cls} \odot S^{ins}; \quad \hat{S}_{i,j,c} = \sum_{k=1}^{U_2} S_{i,j,k,c},$$

Hadamard product

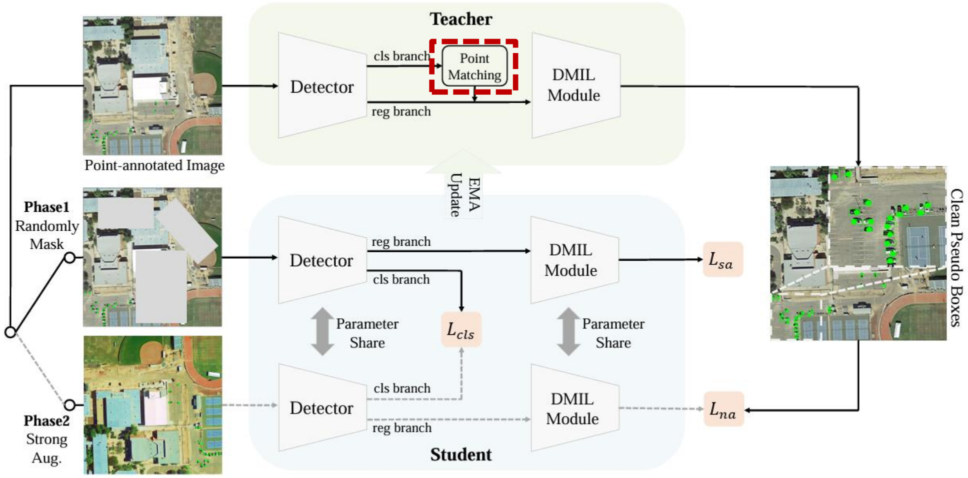
$\hat{S} \in \mathbb{R}^{M \times U_1 \times C}$: Bag Construction에서 구성한 Bag들의 Score

02. Method

3. Noise-aware Label Evolution

- **Point Matching**

2 Stage Top-K Point Matching



$$\mathrm{Cost}(i,j) = \underbrace{(L_{cls}(s_{ji}, c_j))}_{\text{classification cost}} + \underbrace{(1 - \mathbb{1}[p_j \ in \ b_{ji}])}_{\text{spatial cost}},$$

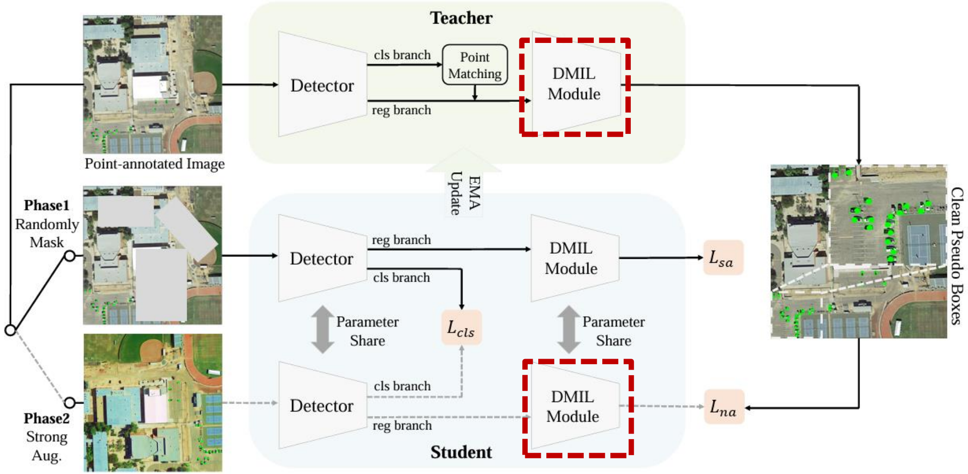
$$\Theta_j = \sum_{k=1}^{K_2} s_{jk} \cdot b_{jk}, \quad where \sum_{k=1}^{K_2} s_{jk} = 1.$$

02. Method

3. Noise-aware Label Evolution

- **Box Refinement**

DMIL Module



$$L_{reg}^{na} = L_{bbox}(P, \Theta')$$
$$L_{reg}^{dmil} = L_{bbox}(B'', \Theta')$$
$$L_{na} = L_{reg}^{na} + \alpha_1 \cdot L_{reg}^{dmil} + \alpha_2 \cdot L_{cls}^{dmil}.$$

02. Method

3. Noise-aware Label Evolution

- **Box Refinement**

Jittering IoU Loss

$$L_{reg}^{na} = \boxed{L_{bbox}(P, \Theta')}$$
$$L_{reg}^{dmil} = \boxed{L_{bbox}(B'', \Theta')}$$
$$L_{na} = L_{reg}^{na} + \alpha_1 \cdot L_{reg}^{dmil} + \alpha_2 \cdot L_{cls}^{dmil}.$$

$$L_{bbox} = L_{IoU}(A, B_{gt}) + \min_i L_{IoU}(A, B'_{gt}[i]).$$

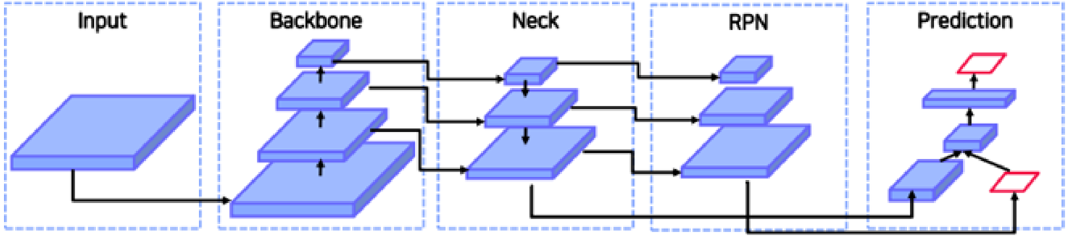
$$B'_{gt} = \begin{cases} (cx_{gt},\ cy_{gt},\ w_{gt},\ h_{gt}) \\ (cx_{gt},\ cy_{gt},\ w_{gt} - w_{gt} \cdot r,\ h_{gt} - h_{gt} \cdot r) \\ (cx_{gt},\ cy_{gt},\ w_{gt} - w_{gt} \cdot r,\ h_{gt} + h_{gt} \cdot r) \\ \cdots \\ (cx_{gt},\ cy_{gt},\ w_{gt} + w_{gt} \cdot r,\ h_{gt} + h_{gt} \cdot r). \end{cases}$$

02. Method

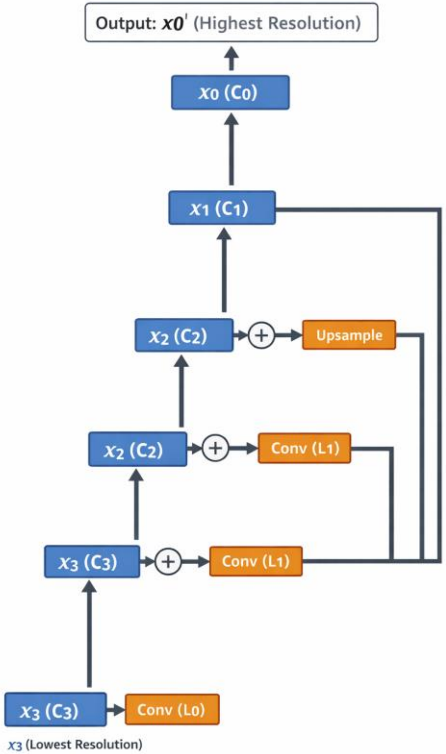
3. Detector Integration

- **Top-down FPN Aggregation**

$$P_{i-1} = P_{i-1} + Up(Conv(P_i)), \quad i \in \{5, .., 7\}$$
$$M = P_{i-1} + Up(Conv(P_i)), \quad i = 4.$$



- **Scale-invariant Label Assignment**
  - **Pseudo Box의 Center Point를 GT Point와 L1 Distance 계산**
  - **가장 가까운 Feature Point를 Positive Sample로 선정**



03

Experiments

03. Experiments

1. Experimental Setup

Tiny Object Detection Dataset

- AI-TOD-v2.0
- TinyPerson, SODA

AI-TOD Dataset

SODA Dataset

- Point Annotation 생성 방안

  - GT Box의 위치에 따라 Point Annotation을 인위적으로 생성

  - Point의 위치는 GT 영역 주변에서 특정 범위 안의 값으로 정의하고, 범위 값은 0~100%까지로 정의

$$px_h = cx + \Delta_x \cdot w \qquad px_o = cx + \Delta_x \cdot w \cdot cos\theta$$
$$py_h = cy + \Delta_y \cdot h \qquad py_o = cy + \Delta_y \cdot h \cdot sin\theta,$$

03. Experiments

1. Experimental Setup

Implementation Details

- ImageNet Pretrained Backbone
- 12 Epoch, SGD, Batch Size : 2
- Metric : AP(@0.25)
- Teacher Model은 Student Model의 EMA Model
- Spatial-aware Box Generation 단계는 학습의 처음에 4,000 Iteration 수행

03. Experiments

2. Main Results

Table 1
Comparison between the Point Teacher and other methods on the AI-TOD-v2.0 `val set` (HBB) and SODA-A (Cheng et al., 2023) `val set` (OBB) with **central annotated points (m = 0%)**. We report $AP_{0.25}$ in this table. Y denotes the end-to-end paradigm. We utilize FCOS as the detector except Point2Rbox-RC (YOLOF). * mean two-stage MIL.

Method	Type	E2E	VE	SH	PE/HE	ST	SP	AI	BR/CT	WM	mAP
HBox-supervised detectors (AI-TOD-v2)											
RetinaNet (Lin et al., 2017c)	HBB	Y	62.0	72.3	26.2	34.8	8.7	2.4	43.6	3.4	31.7
Faster R-CNN (Ren et al., 2017)	HBB	Y	49.0	56.2	19.7	40.6	60.2	38.6	30.9	5.7	37.6
FCOS (Tian et al., 2019)	HBB	Y	75.7	76.2	26.5	65.9	46.2	1.9	36.5	0.3	41.2
Point-supervised detectors (AI-TOD-v2)											
P2BNet (Chen et al., 2022)	HBB	N	0.9	1.9	0.0	7.6	0.1	0.0	1.2	0.0	1.5
P2BNet* (Chen et al., 2022)	HBB	N	0.1	0.8	1.1	16.7	0.0	0.1	0.3	0.0	2.4
PLUG (He et al., 2024)	HBB	N	19.4	51.5	10.0	14.9	54.0	1.2	1.9	0.7	19.2
Point Teacher	HBB	Y	58.2	62.8	17.5	47.1	43.2	1.3	45.4	8.7	**35.5**
RBox-supervised detectors (SODA-A)											
RetinaNet-O (Lin et al., 2017c)	OBB	Y	50.1	82.2	53.0	77.2	88.4	89.0	58.8	87.0	70.7
FCOS-O (Tian et al., 2019)	OBB	Y	61.9	87.4	51.5	81.6	88.6	89.8	60.2	88.7	74.6
Oriented R-CNN (Xie et al., 2021)	OBB	Y	61.0	87.1	56.0	84.9	88.6	89.6	55.5	88.9	74.7
Point-supervised detectors (SODA-A)											
PointOBB* (Luo et al., 2024)	OBB	N	11.7	13.7	56.0	19.7	45.3	80.4	22.8	78.7	37.8
Point2Rbox-RC (Yu et al., 2024)	OBB	Y	6.3	14.4	71.1	48.4	88.8	10.1	3.3	76.0	36.1
PointOBB-v2 (Ren et al., 2024)	OBB	N	26.3	66.4	50.5	55.0	41.4	56.0	44.6	7.2	41.5
Point Teacher	OBB	Y	43.9	77.6	27.5	22.5	85.8	31.8	19.0	72.9	**47.2**

Table 4
Comparison between other methods and our proposed Point Teacher on the TinyPerson (Yu et al., 2020) `test set` with central annotated points. We report $AP_{0.25}$ in this table. We utilize FCOS as the detector.

Method	mAP	AP_{vt}	AP_t	AP_s	AP_m
Hbox-supervised detectors					
RetinaNet	13.9	2.3	7.0	16.1	33.9
Faster R-CNN	15.6	0.0	3.8	19.2	52.2
FCOS	34.2	9.1	24.2	43.5	68.8
Point-supervised detectors					
P2BNet	2.3	0.0	1.0	4.0	8.3
P2BNet*	7.9	0.0	2.4	14.1	24.2
PLUG	7.4	0.0	1.0	8.9	28.1
Ours	**18.6**	3.8	19.3	24.4	29.4

03. Experiments

2. Main Results

Table 2
Comparison between other methods and our proposed Point Teacher on the AI-TOD-v2.0 `val set` (HBB) and SODA-A (Cheng et al., 2023) `val set` (OBB) with **randomly annotated points (m = 100%)**.

Method	Type	E2E	VE	SH	PE/HE	ST	SP	AI	BR/CT	WM	mAP	mAP
HBox-supervised detectors (AI-TOD-v2)												
RetinaNet (Lin et al., 2017c)	HBB	Y	62.0	72.3	26.2	34.8	8.7	2.4	43.6	3.4	31.7	31.7
Faster R-CNN (Ren et al., 2017)	HBB	Y	49.0	56.2	19.7	40.6	60.2	38.6	30.9	5.7	37.6	37.6
FCOS (Tian et al., 2019)	HBB	Y	75.7	76.2	26.5	65.9	46.2	1.9	36.5	0.3	41.2	41.2
Point-supervised detectors (AI-TOD-v2)												
P2BNet (Chen et al., 2022)	HBB	N	0.2	0.5	0.0	1.8	0.0	0.0	0.5	0.0	0.4	1.5
P2BNet* (Chen et al., 2022)	HBB	N	0.6	0.7	0.8	5.5	0.0	0.1	3.4	0.0	1.4	2.4
PLUG (He et al., 2024)	HBB	N	3.8	2.3	2.0	4.1	48.3	3.0	3.0	0.0	8.3	19.2
Point Teacher	HBB	Y	54.5	63.1	16.2	40.4	33.9	0.7	41.9	2.3	**31.6**	**35.5**
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FCOS-O (Tian et al., 2019)	OBB	Y	61.9	87.4	51.5	81.6	88.6	89.8	60.2	88.7	74.6	74.6
Oriented R-CNN (Xie et al., 2021)	OBB	Y	61.0	87.1	56.0	84.9	88.6	89.6	55.5	88.9	74.7	74.7
Point-supervised detectors (SODA-A)												
PointOBB* (Luo et al., 2024)	OBB	N	3.3	2.2	49.4	5.4	65.8	48.6	0.5	72.6	27.9	37.8
Point2Rbox-RC (Yu et al., 2024)	OBB	Y	0.0	0.5	0.1	0.1	7.3	0.0	0.0	1.3	1.0	36.1
PointOBB-v2 (Ren et al., 2024)	OBB	N	10.1	42.8	22.5	43.9	12.4	56.0	8.0	2.2	23.1	41.5
Point Teacher	OBB	Y	16.6	34.2	42.2	62.7	43.8	71.3	13.0	59.3	**40.0**	**47.2**

Table 3
Robustness of other methods and our proposed Point Teache
performance with the point location m set to 0%, 30%, 60%
and 100%. * means two-stage MIL.

m	Point Teacher	PLUG	P2BNet*	P2BNet
0%	35.5	19.2	2.4	1.5
30%	**36.3**	16.7	5.7	1.0
60%	33.0	13.5	3.4	0.6
100%	31.6	8.3	1.4	0.4

03. Experiments

3. Ablation Study

Table 5

Ablations. We train on point-based AI-TOD-v2.0 `train set`, test on `val set`. Phase1 and Phase2 represent Spatial-aware Box Generation phase and Noise-aware Label Evolution phase respectively. Gray row means default setting.

(a) Individual effectiveness of components in Point Teacher.

Phase1	Phase2			mAP
Randomly Mask	Teacher-Student	DMIL	Jittering IoU Loss	
✓				21.0
✓	✓			24.5
✓	✓	✓		34.7
✓	✓	✓	✓	**35.5**

(b) Comparison of DMIL and MIL. * denotes two-stage MIL.

Method	mAP	AP_{vt}	AP_t	AP_s	AP_m
MIL	20.0	10.0	27.6	18.3	17.4
MIL*	20.3	9.7	29.6	15.7	22.9
DMIL	**35.5**	17.5	42.0	37.4	26.6

(c) $\{K_1, K_2, K_3\}$ for Pseudo Boxes Generation.

K_1	K_2	K_3	mAP	AP_{vt}	AP_t	AP_s	AP_m
3	1	1	28.2	25.3	31.8	42.7	0.4
3	1	5	33.6	28.2	37.1	44.3	14.4
5	3	1	**35.5**	17.5	42.0	37.4	26.6
5	3	5	31.9	10.3	47.3	30.1	55.5

(d) r in Jittering IoU Loss.

r	mAP	AP_{vt}	AP_t	AP_s	AP_m
0.0	34.7	12.4	44.5	32.0	21.9
0.2	**35.5**	17.5	42.0	37.4	26.6
0.4	33.2	6.3	56.0	30.2	57.0
0.6	33.0	5.6	44.1	37.6	49.1

(e) β in DMIL Fusion.

β	mAP	AP_{vt}	AP_t	AP_s	AP_m
0.00	34.7	22.6	40.9	46.6	10.9
0.25	**35.5**	17.5	42.0	37.4	26.6
0.50	31.6	13.9	44.0	29.9	58.8
1.00	24.5	5.4	38.8	23.1	32.1

(f) Training proportion of the Spatial-aware Box Generation phase.

Iterations	Proportion	mAP	AP_{vt}	AP_t	AP_s	AP_m
4000	5%	**35.5**	17.5	42.0	37.4	26.6
20000	25%	34.2	5.3	53.4	30.9	51.8
40000	50%	34.1	7.9	50.1	34.0	52.2
80000	100%	21.0	8.0	31.4	18.8	21.9

03. Experiments

3. Ablation Study

Figure 5: Visualization of pseudo boxes generated by the DMIL Module. Green boxes denote the *gt* boxes, yellow boxes denote the pseudo boxes generated by DMIL.

03. Experiments

3. Ablation Study

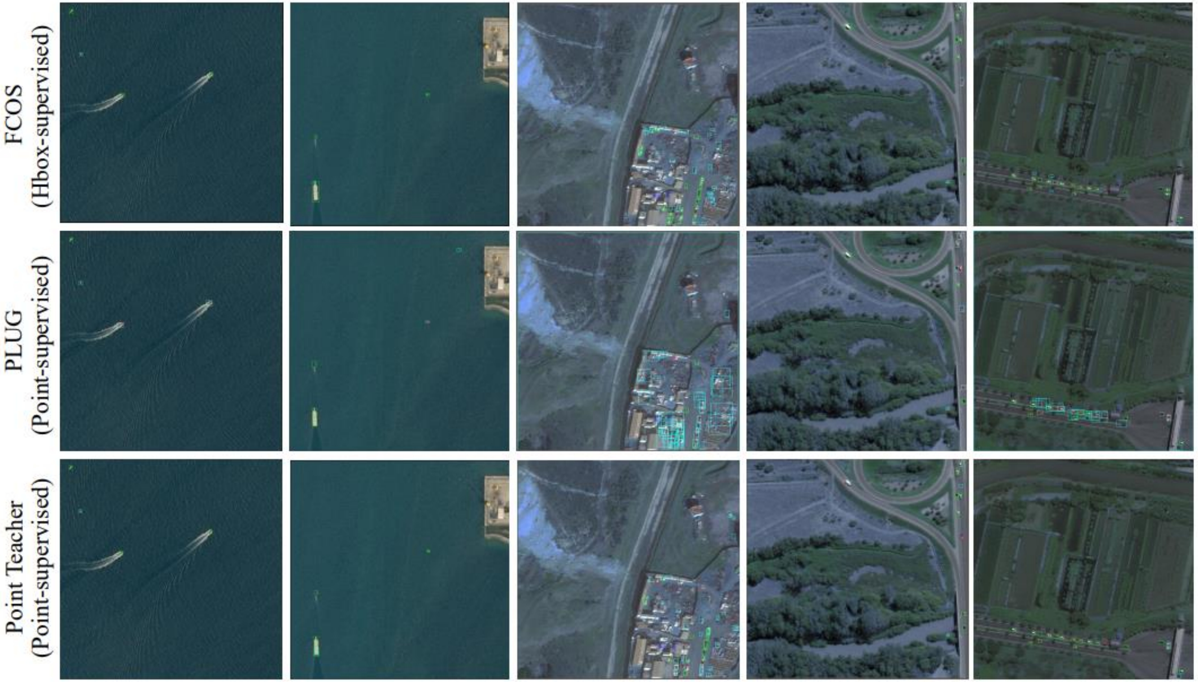


Figure 6: Visualization results on AI-TOD-v2.0 with central annotated points. The first row is the result of FCOS, the second row is the result of PLUG, and the third row is the result of Point Teacher. Green boxes denote true positive predictions, red boxes denote false negative predictions, and blue boxes denote false positive predictions.

04

Conclusion

04. Conclusion

Contribution

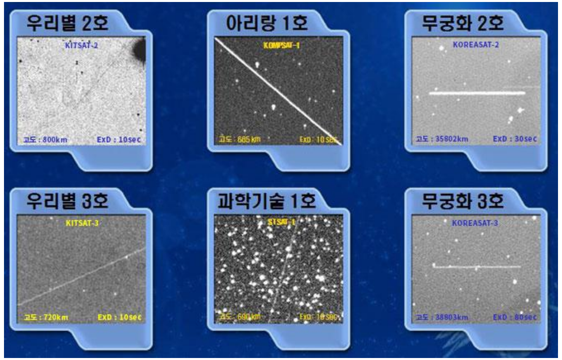
- Cost Effective
- 대규모 Tiny Object Detection Application에서 실용적으로 사용 가능

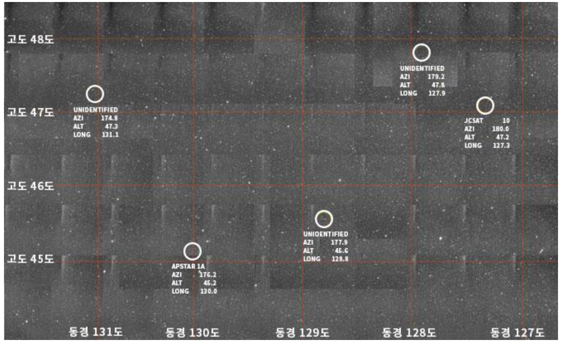
Discussion

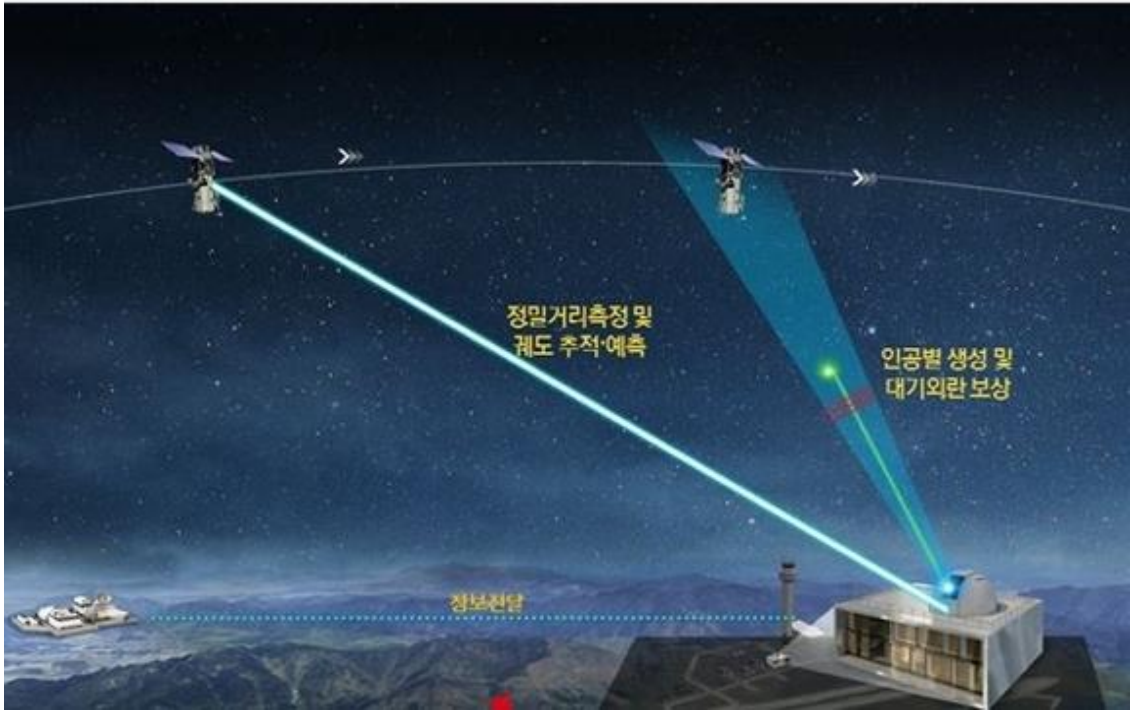
- 왜 기존 Point supervised Method들이 Tiny Object Detection에서 성능이 낮은가
  - MIL Based Method
  - CPM Based Method
  - Auxiliary Based Method
- Limitation of Point Teacher
  - Multi-scale Object
  - Densely packed tiny object

04. Conclusion

2. Real-world Application







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Q & A

Thank you for your attention